Dual-battery dynamic wireless power transfer systems

"IM WORKING ON THE DUAL BATTERY SYSTEMS IN THE DYNAMIC WIRELESS POWER TRANSFER SYSTEMS.SO GIVE ME THE CURRENT EXISTING SOLUTIONS&CURRENT PROGRESS&ALSO THE FEASIBILITY REPORT&THE CONFIDENCE LEVEL OF UR STATEMENTS&ALSO GIVE ME A WELL PLANNED PROPER PATHWAY INCL ALL THE INDUCTIVE AND AC COILS TUNING ALSO HAS TO GAIN KNOWLEDGE ON THE EDDY CURRENTS AND MAGNETIC SHEILDING SO THAT I CAN PROTECT MY OTHER PARTS LIKE ACCELEROMETER MEASURING AND ODEMETER MEASURING UNITS WONT GET MAGENTIC DISTURBANCE" *Generated by LeapSpace*

Dual-Battery Dynamic Wireless Power Transfer Systems at a Glance

Key Findings Table

Topic	Key finding	Practical meaning for your project	Evidence
Existing dual-battery solutions	The supplied evidence does **not** show a clearly documented **experimental dual-battery DWPT vehicle prototype** with explicit pack-to-pack control; the closest evidence is adjacent multifunctional converter or auxiliary-battery concepts, while stronger evidence exists for battery-supercapacitor buffering.	If you want a near-term build, **battery + supercapacitor** is better supported than **battery + battery**.	[1] [2] [3] [4] [5]
Current DWPT progress	Dynamic charging hardware has reached real prototypes, including about **25 kW at 86%**, **100 kW at 80% over 26 cm**, and segmented moving systems with **89.78% peak efficiency** and about **±7.5% output pulsation**.	DWPT itself is technically real, not just theoretical.	[6] [7] [8]

Topic	Key finding	Practical meaning for your project	Evidence
Main technical bottleneck	The dominant cross-cutting issue is **coupling variation** from motion, gap change, and misalignment, which drives power fluctuation, tuning drift, control difficulty, EMI, and battery stress.	Your design should start from misalignment-robust coils and adaptive tuning, not battery architecture alone.	[8] [9] [10] [11] [12]
Coil choice	DD-derived and multi-segment pads generally offer the best robustness trade-off under motion, while self-resonant road coils can lead on raw power density in some tests.	For dynamic vehicle operation, prioritize **misalignment-tolerant geometry** over peak static coupling alone.	[13] [14] [15] [16] [17]
Compensation choice	LCC-family and especially **double-sided LCC** are repeatedly favored for dynamic charging because they stabilize operation under coupling changes and support soft-switching.	Start your first serious DWPT prototype from **DS-LCC / LCC-family**, not plain SS unless you want a simpler baseline.	[11] [18] [19] [20]
Battery protection	Evidence for reduced battery stress exists mainly through indirect hybrid-storage and ripple-reduction studies; direct DWPT battery-aging experiments remain limited.	You can justify buffering and smoothing, but do not overclaim proven life-extension in DWPT yet.	[21] [22] [23] [24] [25] [26]
Shielding	Ferrite-dominant and hybrid passive-plus-active shielding usually preserve coupling better than metal-only shielding; large continuous metal can add eddy-current loss and heating.	Use **ferrite first**, then add optimized metal or active cancellation only where needed.	[27] [28] [29] [30]

Topic	Key finding	Practical meaning for your project	Evidence
Sensor protection	Direct accelerometer/odometer prescriptions are limited, but the evidence supports spacing, orientation control, quiet-zone placement, multilayer shielding with thermal caution, and sensor-level validation/filtering.	Protect sensors by **placement + shielding + filtering**, not shielding alone.	[31] [32] [33] [34]

Direct Answer

For your topic, the most evidence-based answer is this: **DWPT is already technically feasible, but true dual-battery DWPT is still a research gap; today's strongest practical solution is a misalignment-tolerant DWPT link combined with ripple-buffering storage, most credibly battery-plus-supercapacitor rather than battery-plus-battery** [3] [4] [6] [35].

If your goal is to build something robust now, the best pathway is:

1. **Set the storage role first**: main traction battery for energy, fast buffer for transients; current evidence favors supercapacitor buffering more strongly than dual-battery buffering [3] [26].
2. **Choose a dynamic-friendly coupler**: DD-derived, QDS/DSSq-type, or other multi-segment pads when misalignment matters [14] [15].
3. **Use LCC-family compensation**, ideally DS-LCC as your main design branch, with SS only as a simple benchmark [11] [18].
4. **Add adaptive tuning**: frequency tracking, impedance matching, receiver-side or co-tuning, and possibly sub-resonant operation [36] [37] [38].
5. **Smooth battery power** with DC-link buffering or hybrid storage so coil-gap and misalignment events do not directly hit the battery current [22] [39].
6. **Design shielding and conductive layout together** to reduce leakage without creating eddy-current hot spots [29] [30].
7. **Create a sensor quiet zone** for accelerometer and odometer units, then verify with field mapping and sensor-level tests [33] [34].

Bottom-line feasibility judgment:

- **High confidence:** DWPT as a platform is technically feasible [6] [7] [40].
- **Medium confidence:** dual-storage buffering in DWPT is feasible and useful [3] [22].

- **Low confidence:** fully validated on-road **dual-battery** DWPT with explicit pack-to-pack control is already established; the supplied evidence does not support that claim [4] [5].

Study Scope

This report synthesizes the supplied literature on:

- DWPT architectures and energy management, including dual-storage analogs [3] [41]
- Coil geometry, compensation, and tuning for EV wireless charging, largely around the ~85 kHz ecosystem [42] [43]
- Experimental DWPT progress, efficiency, misalignment, and air-gap feasibility [6] [7] [8]
- Eddy currents, shielding, EMC, and sensor disturbance mitigation [29] [33] [44]

Methods in the supplied evidence include reviews, simulations, laboratory prototypes, and real-environment demonstrations [6] [40] [45].

Assumptions & Limitations

- The supplied evidence supports **DWPT feasibility strongly**, but supports **explicit dual-battery DWPT prototypes weakly**; much of the storage evidence comes from battery-supercapacitor or adjacent EV energy-management work rather than direct dual-battery DWPT vehicles [4] [5] [26].
- Direct design rules in centimeters for accelerometer or odometer placement near EV DWPT hardware are **not** provided in the evidence; sensor-protection recommendations are partly indirect [33] [46] [47].
- Some studies report high efficiency under tuned conditions, but deployment-oriented papers stress interoperability, leakage, thermal, and cost barriers; bench efficiency should not be treated as roadway readiness by itself [9] [48] [49].

Suggested Further Research

- Build and publish a **true dual-battery DWPT prototype** with explicit pack-to-pack power-flow logic and battery-stress metrics [4] [5].
- Compare **battery + battery** versus **battery + supercapacitor** buffering under segment gaps, misalignment, and transient power interruptions [50] [51].
- Perform direct **battery-aging tests** under DWPT ripple and coupling variation [24] [25].
- Quantify **sensor immunity** for accelerometers, wheel-speed, Hall, and odometer modules near ~85 kHz charging hardware [33] [47].
- Co-optimize **coil geometry, compensation, shielding, and storage buffering** in one validation loop instead of treating them separately [11] [14] [30].

Full Report Body

1. Introduction and Scope

Dual-battery DWPT aims to combine **in-motion wireless charging** with **onboard energy buffering or energy sharing between storage units** so the vehicle can reduce range anxiety, smooth charging power, and potentially reduce stress on the main traction battery [9] [21]. The supplied literature strongly supports DWPT as a route to extend range or reduce required onboard battery size, but it also shows that stable operation depends on the combined behavior of couplers, compensation, converters, and control, not on storage architecture alone [9] [10] [52].

A key practical point for your project is that **sensor protection and EMC are not secondary issues**. Leakage fields, conductive structures, and shielding choices interact with efficiency, thermal loss, and nearby electronics behavior [29] [44] [53]. That matters directly for accelerometer and odometer-related units.

Confidence level: High - The scope and system-level constraints are repeated across multiple reviews and prototypes [9] [40] [54].

2. Foundational Context

2.1 What matters most in DWPT

DWPT performance is governed by changing **magnetic coupling**, which varies with lateral offset, air-gap variation, and vehicle motion [9] [10]. That coupling variation changes reflected impedance, output power, and the operating point of the resonant network, which then affects efficiency, control stability, and battery-side stress [10] [11] [55].

2.2 Why dual storage is discussed

Wireless charging literature repeatedly notes that battery management and WPT are often treated separately, even though integration could improve charging efficiency and balancing [41] [56]. In practice, the strongest available evidence does **not** yet show mature dual-battery DWPT hardware, so the field often uses **dual-storage analogs**, especially battery-supercapacitor systems, to absorb fast transients while the battery handles slower average energy flow [3] [26].

2.3 Why coils and compensation dominate outcomes

The literature repeatedly identifies three core design levers: **coil geometry**, **compensation network**, and **converter/control strategy** [9] [10] [42] [57]. Under dynamic conditions, these dominate whether your power transfer stays stable enough for any battery strategy to work [8] [11].

Confidence level: High - This system view is well supported across architecture, coil, and safety sources [9].

3. Current Existing Solutions for Dual-Battery DWPT

3.1 Observed architectures

The evidence set does **not** provide a clearly documented experimental dual-battery DWPT vehicle prototype with full battery-to-battery control logic [4] [5]. The closest reported architecture is a multifunctional converter in which the traction battery can charge an auxiliary battery through a DC-DC path while supporting other operating modes [1]. A simulation-only concept also uses a low-voltage auxiliary battery to charge the traction battery for emergency range extension when traction SOC drops below a threshold [2].

By contrast, the strongest storage-buffering evidence near DWPT uses **battery-supercapacitor hybrids**. These route fast transients, pulsations, and regenerative spikes to the supercapacitor while the battery sees slower energy flow [3] [35]. Taken together, these findings suggest that the current "existing solution" closest to your need is better described as **dual-storage DWPT** rather than mature **dual-battery DWPT** [3] [4].

3.2 Energy-management strategies

The supplied papers report fuzzy logic, adaptive fuzzy logic, rule-based control, finite-state-machine control, MPC, and state-machine logic for hybrid or dual-storage EV power management [58] [59] [60] [61] [62] [63] [64]. These controllers use SOC, load demand, road conditions, slope, or voltage state to switch modes or split power [60] [62] [63] [65].

However, the reported metrics are mostly qualitative: better efficiency, lower battery stress, improved voltage regulation, and lower current loss or RMS current [59] [61] [64] [66]. The supplied content does not provide a strong measured benchmark for charging ripple specifically in **dual-battery wireless-charging vehicles** [67].

3.3 What is actually mature today

The mature part is **DWPT power transfer**, not dual-battery coordination [6] [7]. Therefore, for a practical project, current solutions should be ranked as:

1. **Most evidence-backed:** DWPT + battery-supercapacitor buffering [3] [35].
2. **Partially evidenced:** multifunctional converter paths between traction and auxiliary subsystems [1].
3. **Weakly evidenced:** full experimental dual-battery DWPT with pack-to-pack real-time control [4] [5].

Confidence level: Medium-High - Strong on what is absent and what is well supported; weaker on specific dual-battery implementations because the direct evidence is sparse [4] [5].

4. Current Progress and Research Trends

4.1 Hardware progress

DWPT hardware progress is substantial. A real-environment 25 kW system achieved 86% efficiency and delivered equal energy across ±15 cm lateral misalignment [6]. A roadway-powered moving-EV magnetic-resonance system reached 100 kW at 80% efficiency across a 26 cm air gap [7]. A segmented dynamic WPT prototype achieved 89.78% peak efficiency with about ±7.5% output pulsation [8].

These are important because they show that **moving-vehicle charging, large air gaps, and segment-based transfer are already feasible at meaningful power** [6] [7] [8].

4.2 Progress in buffering and battery protection

Battery-side pulsation is recognized as harmful to lifetime and efficiency, which motivates output stabilization and buffering [21] [55]. Experimental mitigation includes a hybrid LCC-S and S-LCC topology with bipolar pads that limited output fluctuation to within ±10% across speed and misalignment changes [39]. Another experiment used a battery-plus-supercapacitor hybrid with an exponentially filtered battery power profile to compare aging against a battery-only case while keeping extracted energy constant [22].

Direct DWPT battery-aging evidence remains limited, but related evidence links current ripple and current gradients to faster degradation, including about 50% higher degradation rate for higher current gradients versus constant-current operation [24] [25].

4.3 Standards-driven direction

Research is increasingly shaped by the compliance ecosystem around EV wireless charging near ~85 kHz, with interoperability and exposure compliance acting as design constraints [42] [43] [44]. That pushes designs toward more compliance-aware coil shapes, shielding, and field control rather than raw efficiency only [68].

Confidence level: High - Prototype progress is directly evidenced; standards influence is consistent though less detailed in deployment practice [6] [43].

5. Feasibility Report

5.1 Technical feasibility

DWPT is technically feasible for real vehicles. Reviews and demonstrations show meaningful power, tolerable air gaps, and workable misalignment handling [6] [7] [40]. Efficiency envelopes in the supplied studies range from modest to high depending on speed, gap, and tuning, with reports spanning 55% to above 90% in different systems and conditions [69] [70] [71].

5.2 Practical barriers

The main barriers are repeated across the literature: misalignment, coupling fluctuation, leakage field compliance, thermal limits, interoperability, foreign-object detection, and cost/infrastructure complexity [9] [43] [54] [72] [73]. One prototype study notes operation designed down to coupling 0.15, below which primary resonant current may exceed thermal limits of resonant components and inverter switches [74]. Reviews also report that a 5 cm pad displacement can reduce power by 20-30% and increase leakage by 40-60% in some cases, even though other systems maintain >85% efficiency across ±150 mm misalignment [73].

5.3 Feasibility of dual-battery specifically

Here the evidence is much weaker. The supplied research does not verify a hardware-prototype dual-battery DWPT vehicle with explicit mode switching, battery-stress metrics, or pack-to-pack power control [4] [5]. Therefore:

- **DWPT platform feasibility:** high [6] [40]
- **Dual-storage buffering feasibility:** medium-high [3] [22]
- **Dual-battery DWPT maturity:** low [4] [5]

5.4 Overall judgment

Taken together, the evidence implies that your project is feasible **if framed as a co-design problem**: storage buffering, coupler robustness, compensation stability, and shielding must be designed together [8] [11] [30]. A thesis or prototype focused on a **main battery + smaller buffer storage path** is more aligned with current evidence than claiming an already mature dual-battery solution [3] [26].

Confidence level: High for DWPT, Medium for your exact dual-battery target - The distinction comes directly from the evidence base [4] [6].

6. Coil Design and Resonant Tuning Pathway

6.1 Coil geometry selection

Simple circular and square pads can work, but dynamic systems reward geometries that flatten performance under misalignment rather than maximize only peak alignment performance [13] [14]. DD pads are directional and asymmetric, so DD-derived compound geometries such as QDS, DSSq, orthogonal double-DD, and triangulated or multi-coil layouts are used to broaden tolerance [14] [15] [75] [76].

For raw roadway power, one 85 kHz self-resonant coil delivered about 30 kW, around 8 kW more than circular and DD in the cited test [13]. For robustness under motion, the broader trend favors DD-derived multi-segment transmitters [14] [15].

6.2 Compensation network selection

SS, SP, PS, and PP are foundational topologies, with SS attractive for simplicity and constant-current behavior in ideal conditions [77] [78]. But in dynamic EV charging, SS/SP are more sensitive to coupling drop and misalignment [12] [79].

LCC-family networks are more suitable for dynamic work. LCC-LCC can provide approximate constant current, LCC-S can provide load-independent primary current and ZVS, and DS-LCC is widely adopted because it stabilizes resonance under coupling fluctuation and lowers output-voltage sensitivity [11] [18] [80].

6.3 Adaptive tuning

Adaptive frequency tuning, impedance matching, and co-tuning of transmitter and receiver are used to maintain resonance as coupling changes [36] [81] [82]. A specific sub-resonant control result reports that a 2.5% resonant-frequency increase widened the upper coupling bound by 13.9% and the lower bound by 2.2% with less than 0.3% efficiency loss [37]. A 3.3 kW, 85 kHz EV charging system improved efficiency by 4.67% under misalignment through dynamic adjustment of coupling coefficient and frequency [38].

6.4 Practical tuning pathway for you

1. Build an **SS baseline** to understand reflected impedance and resonance behavior [77].
2. Move to **DS-LCC** for your serious dynamic platform [11].
3. Characterize coupling versus lateral offset, air gap, and segment gap [8] [9].
4. Add **adaptive frequency or impedance tuning** once the passive design is stable [36] [37].
5. Only then integrate storage-side buffering and EMS [22] [60].

Confidence level: High - Compensation and tuning conclusions are strongly repeated across sources [11] [20] [37].

7. Eddy Currents, Magnetic Shielding, and Sensor Protection

7.1 Eddy-current mechanisms

Time-varying fields induce circulating currents in nearby conductive parts, causing I^2R heating and efficiency loss; these losses depend on penetration depth, skin depth, frequency, resistivity, thickness, and geometry [29] [83] [84] [85]. Reported hot spots include chassis parts, steel rods, aluminum endplates, and metallic shields or panels near the active region [29] [86] [87].

Mitigation strategies include non-conductive supports, insulating conductive supports from laminations, segmented or laminated cores, ferrite guidance, and layout changes that reduce field penetration into conductors [29] [85] [88].

7.2 Shielding options and trade-offs

Ferrite-dominant shielding improves flux confinement and generally preserves coupling better than metal-only backing [28] [89] [90]. Aluminum can reduce leakage, but unless optimized or combined with ferrite it can hurt coupling and create eddy-current loss [91] [92] [93].

The strongest trade-offs are reported for **hybrid passive-plus-active** or **active-shielded** systems. Active shielding coils reduced leakage by 43.63% while maintaining 95.58% efficiency in one study, and another active shield achieved 86.97% leakage reduction with only 0.28% efficiency loss [27] [94]. Hybrid passive/reactive shields also showed strong leakage reduction with efficiency above 90% [95] [96].

7.3 Protecting accelerometer and odometer units

Direct EV-DWPT prescriptions for accelerometers and odometers are limited, but the evidence supports a layered strategy:

- **Place sensors away from strong leakage lobes** rather than directly above or below the coupler [34].
- **Control orientation** where magnetic axis sensitivity matters [97].
- Use **ferrite-dominant or combined conductive + magnetic shielding** where space allows [34] [98].
- Be cautious with **multilayer wraps** around accelerometer-related hardware because they can impair heat dissipation and create temperature-driven measurement errors [33].
- Add **software-side filtering or covariance weighting** so disturbed measurements are down-weighted instead of blindly trusted [32].
- Validate with **field mapping plus sensor output tests**, not field mapping alone [33] [44].

For wheel/odometer-related sensing, the supplied evidence notes that larger air gaps can ease positioning constraints and that AMR sensors can use larger gaps than Hall sensors in some contexts [46] [99]. That suggests a practical inference: if you must place speed-sensing hardware near charging hardware, prefer placement and sensing arrangements tolerant to larger gap and lower stray coupling. **Inferred: low confidence** [46] [99].

Confidence level: Medium-High - Strong for shielding and eddy-current physics; medium for direct accelerometer/odometer prescriptions because evidence is indirect [27] [29] [33].

8. Safety, EMC, and Standards

ICNIRP-based exposure limits are the main human-exposure references in the supplied studies, and EV WPT work around 85 kHz commonly checks against such limits or related international safety guidance [43] [68] [100] [101]. SAE J2954 is identified in the evidence as a wireless-charging standard for interoperability, EMC, and EMF safety, and one study reports reference-level exceedance under worst-case conditions before mitigation [43].

Compliance assessment in the supplied papers uses FEM, FDTD, method of moments, impedance methods, and measured magnetic-field data [44] [45]. One paper supports three-point magnetic-field measurement at 0.5, 1.0, and 1.5 m for safety compliance [45].

For your work, EMC should be treated as a **design input**, not an end-of-project test [9] [44].

Confidence level: High - Safety/compliance methods are directly evidenced [43] [45].

9. Recommended Development Pathway

Phase 1: Foundations

- Learn coupling coefficient behavior, reflected impedance, resonance, and compensation behavior under misalignment [11] [102].
- Study EV wireless charging standards context and exposure constraints early [42] [43].

Phase 2: First hardware branch

- Build a simple **SS baseline** with circular or square coils to understand measurements and losses [42] [77].
- Measure coupling versus offset and air gap [9] [10].

Phase 3: Dynamic-ready branch

- Move to **DD-derived or multi-segment pads** for motion robustness [14] [15].
- Shift to **DS-LCC or LCC-family compensation** [11] [18].
- Add segmented transmitter logic if dynamic continuity matters [8].

Phase 4: Storage buffering

- Start with **battery + supercapacitor** because it is the most evidence-backed transient buffer [3] [35].
- If pursuing novelty, then test **main pack + small buffer pack** as a research contribution, but label it experimental [4] [5].

Phase 5: Tuning and control

- Implement adaptive frequency or impedance tuning [36] [82].
- Compare fixed-frequency and sub-resonant control [37].
- Add EMS only after passive power-stage stability is verified [60].

Phase 6: Shielding and sensor immunity

- Map leakage fields first [44].
- Add ferrite shielding, then optimize metal or active shielding only if needed [27] [28].
- Keep conductive mechanical parts out of strong flux paths to reduce eddy-current hot spots [29] [30].

- Create a validated sensor quiet zone and combine shielding with filtering [32] [33].

Phase 7: Validation

- Efficiency versus offset, gap, speed, and segment transitions [6] [8].
- Thermal checks at low coupling and worst-case alignment [74].
- EMC and field exposure checks using simulation plus measurement [44] [45].
- Sensor disturbance tests on accelerometer and odometer assemblies during charging operation [33] [34].

Confidence level: High - This pathway is directly assembled from repeated technical patterns in the evidence [8] [11] [29].

10. Final Conclusion

The evidence supports a clear conclusion: **DWPT is real and increasingly capable, but explicit dual-battery DWPT remains underdeveloped experimentally** [4] [6]. The most defensible engineering path today is to treat your problem as a **co-design of dynamic coupler robustness, LCC-family compensation, adaptive tuning, transient buffering, and EMC-aware shielding** [11] [14] [30] [38].

If you want a practical system now, choose **DD-derived/multi-segment coils + DS-LCC + adaptive tuning + battery/supercapacitor buffer + ferrite-dominant shielding + validated sensor quiet zone** [3] [11] [14] [28] [33]. If you want a research contribution, the strongest gap is a **true dual-battery DWPT prototype** with explicit pack-to-pack power-flow control, battery-stress metrics, and sensor-immunity validation under motion [4] [5] [33].

Overall conclusion confidence: Medium-High - High for DWPT feasibility and design priorities, lower for dual-battery-specific maturity because direct prototype evidence is missing [4] [6].

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